

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Write and then solve an equation that represents the statement: Fourteen less than six times a number, n , is 64.

Answer: $6n - 14 = 64 \rightarrow 6n = 78 \rightarrow n = 13$

2. Solve: $8x + 21 = 93$

Answer: $8x = 72 \rightarrow x = 9$

3. Solve: $6(2x + 5) - 8 = 46$

Answer: $12x + 30 - 8 = 46 \rightarrow 12x = 24 \rightarrow x = 2$

4. Solve: $7(x - 2) = 4x + 10$

Answer: $7x - 14 = 4x + 10 \rightarrow 3x = 24 \rightarrow x = 8$

5. Solve: $(3/5)x + 4 = 22$

Answer: $(3/5)x = 18 \rightarrow x = 30$

6. Solve: $0.8x - 2.4 = 5.6$

Answer: $0.8x = 8 \rightarrow x = 10$

7. Solve each equation. If it has no solution or is an identity, say so and explain how you know.

a. $9x - 6 = 9x + 5$

Answer: no solution — $-6 \neq 5$

b. $5(2x - 7) = 10x - 35$

Answer: identity — true for all values of x

Algebra 1 – Module 2 Practice Test •

WORKED KEY

8. Solve the literal equation $P = 2a + b$ for a .

Answer: $a = (P - b)/2$

9. Solve the literal equation $V = (1/3)\pi r^2 h$ for h .

Answer: $h = 3V/(\pi r^2)$

10. A bowling alley charges a \$4.50 shoe rental plus \$5.25 per game. Talia spends \$30.75 in all. Write and solve an equation to find how many games, g , she bowled.

Answer: $5.25g + 4.5 = 30.75 \rightarrow 5.25g = 26.25 \rightarrow g = 5$ games

11. Error Analysis — To solve $5(x + 4) = 45$, Rex first wrote $5x + 4 = 45$. Identify his mistake, then solve it correctly.

Answer: he distributed the 5 to the x but NOT to the 4; correct: $5x + 20 = 45 \rightarrow x = 5$

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12. The sum of three consecutive integers is 111. Find the integers.

Answer: $n + (n+1) + (n+2) = 111 \rightarrow 3n + 3 = 111 \rightarrow n = 36$; 36, 37, 38

13. A rectangle's length is 3 cm more than twice its width, and its perimeter is 78 cm. Write and solve an equation to find the width and the length.

Answer: $2w + 2(2w + 3) = 78 \rightarrow 6w + 6 = 78 \rightarrow w = 12$ cm, $l = 27$ cm

Cumulative Review — ACT Style. Circle the best answer.

14. What is the value of $5x^2 - 8$ when $x = 3$?

(A) 22
(B) **37**
(C) 53
(D) 217

Answer: $5(9) - 8 = 37$

15. Which expression is equivalent to $7(2x - 3) + 5x$?

(E) **$19x - 21$**
(F) $19x - 3$
(G) $9x - 21$
(H) $19x + 21$

Answer: $14x - 21 + 5x = 19x - 21$

16. 40% of 85 is what number?

(A) 30
(B) **34**
(C) 45
(D) 51

Answer: $0.40(85) = 34$